

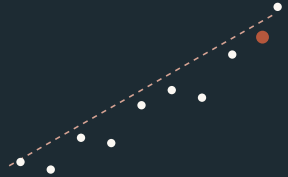
AN INTRODUCTION FOR UNDERGRADUATES

Data Science

Lecture 1 — From Data to Decisions

Md. Owaliur Rahman

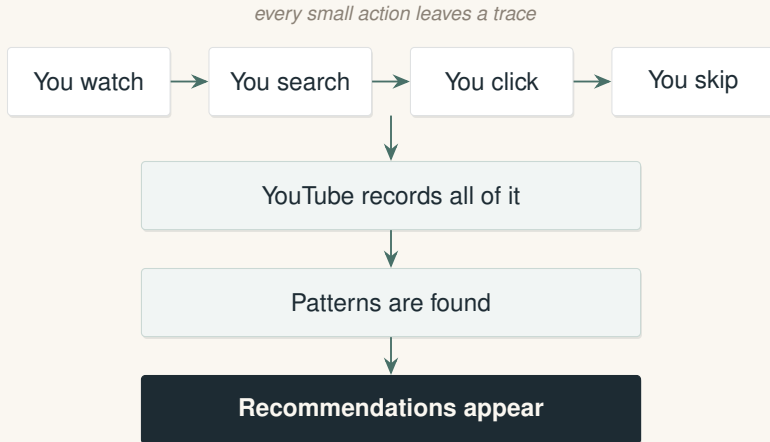
PRESIDENCY UNIVERSITY



A QUESTION TO BEGIN WITH

How does YouTube know
what you might want
to watch next?

What actually happens behind the screen



Three ordinary examples

Student marks

78 65 91
54 83

Weather

32°C
78% humidity
Rain = Yes

Online activity

Video watched
Video skipped
Video liked

What do these three have in common?

Data

Data is a collection of recorded
**facts, observations, or
measurements.**

a number

a word

a yes / no

a click

a photograph

all of these are data

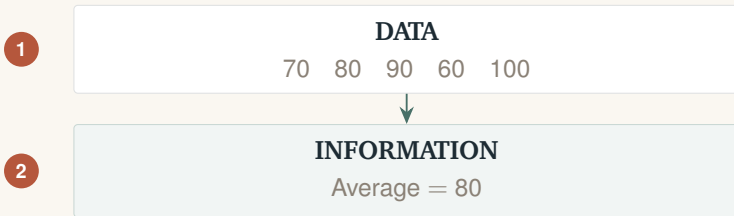
Is raw data useful by itself?

1

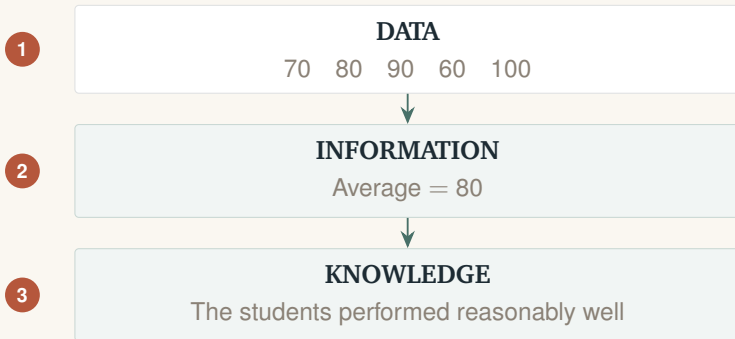
DATA

70 80 90 60 100

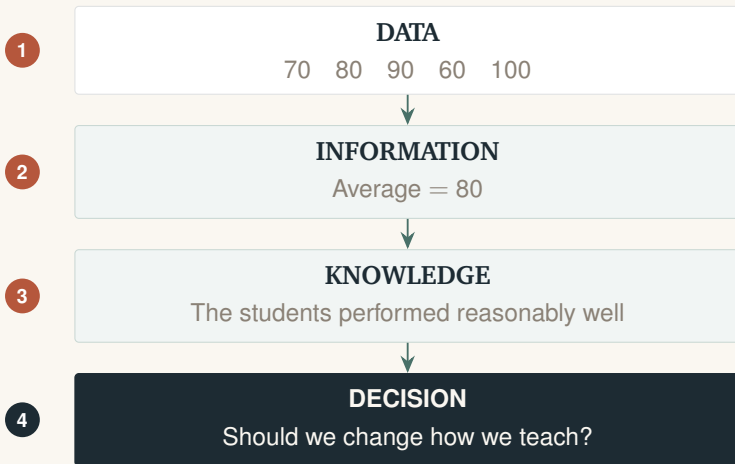
Is raw data useful by itself?



Is raw data useful by itself?

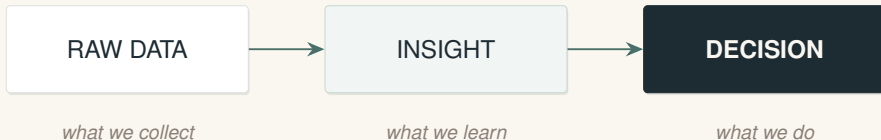


Is raw data useful by itself?



The key idea

Data becomes useful only when we
extract meaning from it.



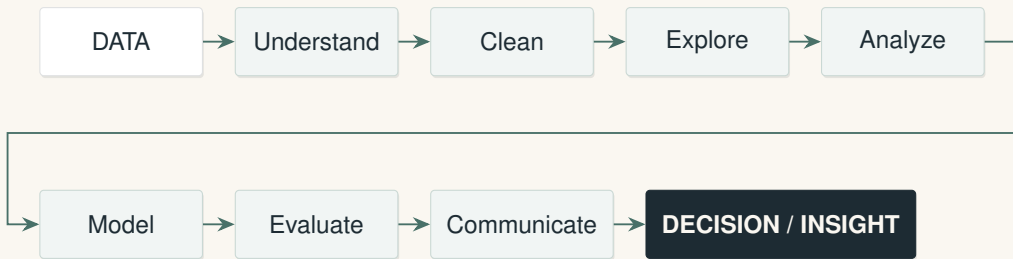
SO, WHAT IS DATA SCIENCE?



Data Science is the process of
extracting useful insights and knowledge
from data.

It is a *process* — not one tool, not one algorithm.

Data Science is a process



Every box is a real job. We will spend the whole course inside them.

A real problem

A university wants to identify students who **may struggle academically**, early enough to help.

What data might help us?

A real problem

A university wants to identify students who **may struggle academically**, early enough to help.

What data might help us?

Attendance

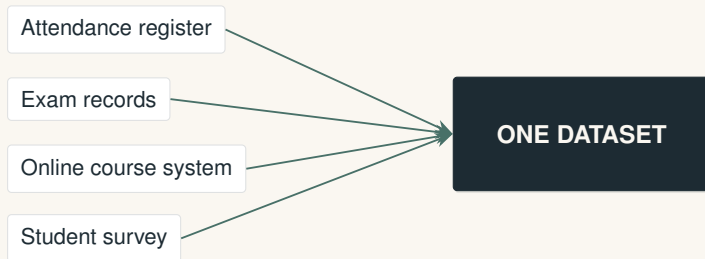
Previous GPA

Assignment marks

Study hours

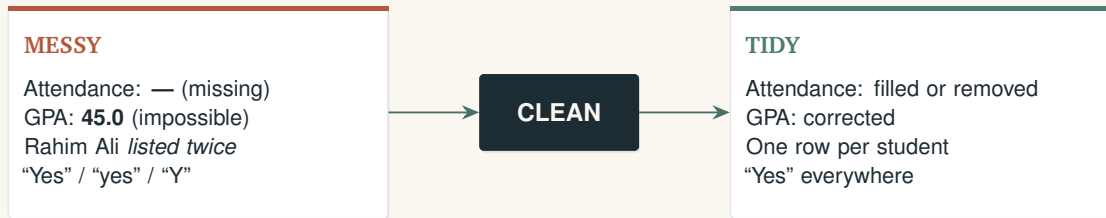
Course load

Collect the data



four offices, four formats, one student

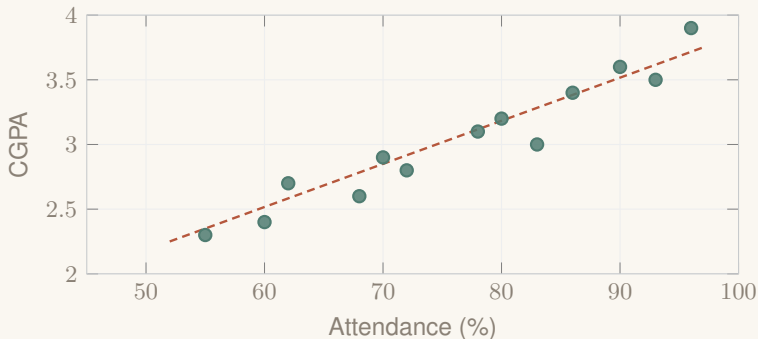
Clean the data



Real data is never neat. This step often takes the most time.

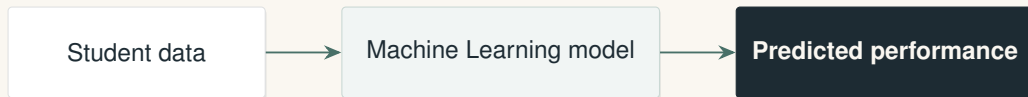
Explore the data

Does attendance relate to performance?



Each dot is one student. We are only *looking* — no conclusions yet.

Build a model

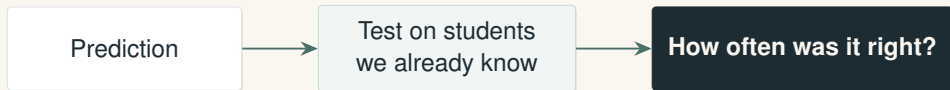


*it learns patterns from students
whose results we already know*

A model learns patterns from **past examples**
and applies them to **new cases**.

Evaluate

Can we trust the prediction?



A model can be wrong — and being wrong has a cost.

Communicate the result



A Data Scientist does **not** stop
at building a model.

If nobody understands the result, nothing changes.

Six familiar places

YouTube / Netflix

recommendations

Banking

fraud detection

Healthcare

pattern discovery

E-commerce

product suggestions

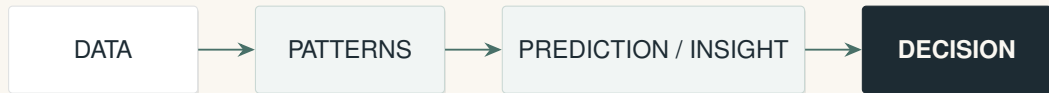
Transportation

routes and schedules

Education

student performance

The pattern underneath all of them



The application changes. The idea does not.

Each field asks its own question

Programming

What can we make a computer do?

Statistics

What can we learn from data?

Machine Learning

Can a computer learn patterns from examples?

Data Science

How do we get useful knowledge from data?

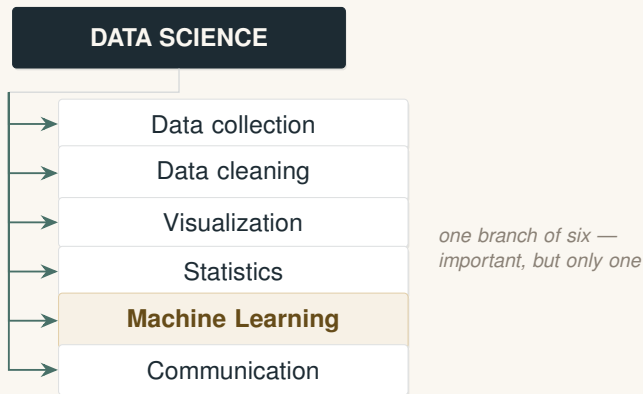
Artificial Intelligence

Can machines do things that need intelligence?

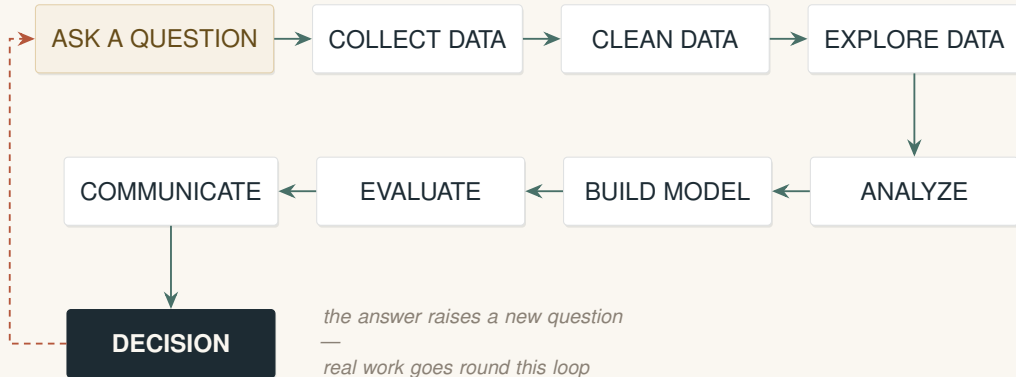
These are not separate worlds — they overlap heavily.

A common misconception

DATA SCIENCE $\neq$ MACHINE LEARNING



The complete Data Science workflow



How this course follows the workflow

- **Lectures 1–2** Understanding data
- **Lectures 3–5** Python and handling data
- **Lectures 6–8** Visualization, statistics, exploration
- **Lectures 9–11** Machine learning and evaluation
- **Lecture 12** One complete end-to-end project

Nothing here assumes you already know Python or statistics.

Look at this data

Student	Study hours	Attendance	CGPA
A	2	60%	2.4
B	5	80%	3.2
C	7	90%	3.7
D	3	70%	2.8
E	8	95%	3.9

1. What data
do we have?

2. What patterns
can you see?

3. What question
could we ask?

Questions this data lets us ask

1

Does study time relate to CGPA?

2

Does attendance relate to CGPA?

3

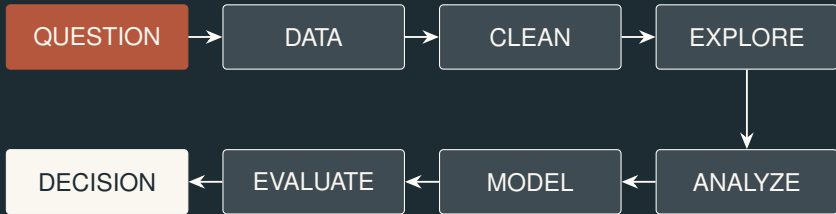
Can we **predict** CGPA from these two things?

The first two are *exploration*. The third needs a *model*.

Five things to remember

- 1 Data is recorded facts and observations.
- 2 Data Science is extracting useful knowledge from data.
- 3 Data Science is much more than machine learning.
- 4 Real-world data is messy.
- 5 Data Science turns data into insights and decisions.

THE JOURNEY OF THIS COURSE



That is the journey we will learn
in this course.

A small thing to bring next class

Pick one recommendation you received today — a YouTube video, an Instagram Reel, a TikTok, a Daraz product, a Netflix movie, a Facebook post, a Spotify song. Then write:

- 1 What was **recommended**?
- 2 **Why** do you think it was recommended to you?
- 3 What **data** about you might have been used?
- 4 What other recommendation would you expect **next**?

Questions you should be able to answer

DATA AND DATA SCIENCE

- 1 What is data? Give three examples that are not numbers.
- 2 Explain data → information → knowledge → decision.
- 3 Define Data Science. Why is it a *process*?
- 4 Why is Data Science $\neq$ Machine Learning? Name its six parts.
- 5 Statistics, ML, AI, Data Science — what does each one ask?

THE WORKFLOW

- 6 List the stages of the workflow, in order.
- 7 What is data cleaning? Name four problems it fixes.
- 8 Why does cleaning usually take the most time?

Questions that ask you to apply it

MODEL, EVALUATE, COMMUNICATE

- 9 Why is communicating the result part of the job?
- 10 A scatter rises to the right. What may we *not* say?

USE IT ON SOMETHING REAL

- 11 Explain a YouTube recommendation in four stages.
- 12 Name four fields that use Data Science, with one use each.
- 13 Study hours, attendance, CGPA — what patterns?
- 14 Which of those need exploration, which need a model?